

Final Project Proposal

FairHire AI: Mitigating Bias in Hiring Models

Group Members

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Problem Statement & Motivation

Artificial Intelligence (AI) and Machine Learning (ML) are increasingly used in recruitment to automate resume screening and candidate selection. While these systems improve efficiency, they can inherit and amplify biases in historical hiring data, leading to discriminatory outcomes based on attributes like gender, ethnicity, or socioeconomic background. Such bias raises serious ethical, legal, and societal concerns. Real-world examples show that even high-performing models can produce unfair outcomes.

This project investigates the research question:

To what extent do machine learning models exhibit measurable bias in hiring decisions and how effectively can mitigation strategies reduce this bias without significantly impacting performance?

We will use a hiring discrimination dataset containing candidate features and outcomes, enabling analysis of intersectional bias (gender and ethnicity). The goal is to quantify bias in baseline models, apply mitigation techniques, and evaluate the trade-offs between fairness and accuracy to help design fairer AI-driven hiring systems.

Approach

Task: Binary classification (hire vs. no hire)

Preprocessing: Clean data, encode categorical features, normalize, identify protected attributes

Models: Logistic Regression, Decision Tree, Random Forest

Fairness Metrics:

- Demographic Parity
- Equal Opportunity

Mitigation:

- Pre: Reweighting
- In: Fairness constraints
- Post: Threshold adjustment

Tools: Python, scikit-learn, Fairlearn/AIF360, Pandas, NumPy, Matplotlib

Related Work & Novelty

Prior work shows ML can replicate bias and highlights fairness-accuracy trade-offs. Studies such as Barocas et al. (Fairness in Machine Learning) and Kleinberg et al. (Inherent Trade-Offs in Fair Determination) highlight the challenges of balancing fairness and accuracy. Tools such as IBM's AIF360 provide frameworks for measuring and mitigating bias. Our work evaluates multiple fairness metrics, studies intersectional bias, and compares mitigation methods across pipeline stages. Our contribution extends prior work by evaluating multiple fairness metrics, analyzing intersectional bias, comparing pre-, in-, and post-processing mitigation techniques, and examining trade-offs between fairness and predictive performance. This provides a more comprehensive evaluation of fairness in AI-based hiring systems.

Evaluation Approach

Performance Metrics: Accuracy, Precision, Recall, F1-score

Fairness Metrics: Demographic Parity Difference, Equal Opportunity Difference

Comparison Strategy: Compare baseline models with models using bias mitigation techniques, using tables and visualizations to show fairness–performance trade-offs.

Success Criteria: Reduce bias significantly while maintaining acceptable performance.

Milestones

Weeks 1–2: Data + literature

Week 3: Models

Week 4: Bias measurement

Weeks 5–6: Mitigation

Week 7: Evaluation

Week 8: Final report

Division:

Sravya: Data + models

Mohammad: Fairness + mitigation

Both: Analysis + report

Bibliography

- Barocas, S., Hardt, M., & Narayanan, A. *Fairness in Machine Learning*
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). *Inherent Trade-Offs in the Fair Determination of Risk Scores*
- IBM Research. *AI Fairness 360 (AIF360) Toolkit*
- Dataset: Buttler, Dominik (2025). *Hiring Discrimination in the Organisational Context*

